Research Presentation

Data-Driven Corporate Credit Assessment

Machine Learning Approaches to Credit Rating Prediction

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Motivation and Introduction

Motivation

Information Asymmetry

Reducing information gaps between borrowers and lenders

Capital Requirements

Managing risks and meeting capital regulations

Nonlinear Patterns

Traditional models miss nonlinear patterns

Primary Objective

Utilize hybrid machine learning models (cluster-then-predict) to improve credit rating prediction accuracy and interpretability

Multiple ML Models		Clustering
Accuracy	x	**Interpretability**
Data Transformation		Feature Reductions

Proposed Hybrid Solution

To balance out between the accuracy (model complexity) and interpretability (model simplicity)

Structured Framework

We propose a structured predictive framework that integrates **high-quality financial data** with rigorous **data preprocessing**, **transformation**, and **feature selection**

Clustering Integration

Our model utilizes **clustering** to identify latent data structures before **classification**, helping to handle noise and inconsistencies before prediction.

Data and Benchmark Method

Data Overview: TEJ & TCRI

Data sourced from the Taiwan Economic Journal (TEJ) using Taiwan Corporate Rating Index (TCRI) ratings as the response variable.

21,676 observations

48

Candidate Predictors

6,622 observations

10

Selected Predictors

Feature selection performed via forward and backward stepwise methods

Data Exploration

Table 2: Summary of missing values in predictors

Variable	Missing Values	Missing (%)
Expected credit impairment (loss) profit operating expense	2,437	36.80
Retention ratio	1,455	21.97
Operating leverage	956	14.43
Financial leverage	845	12.76
R&D expense	232	3.50
Interest coverage ratio	191	2.88
Cash flow adequacy ratio	107	1.62
Cash reinvestment ratio	36	0.54
Contingent liabilities to net worth	2	0.03
ROE comprehensive income	1	0.02
ROA comprehensive income	1	0.02

Table 3: Descriptive statistics of selected predictors

Variable	Count	Mean	Std. Dev.	Min	25%	Median	75%	Max	Skewness	Kurtosis
Total liabilities	6622	11552982.53	62382019.86	4946.00	944673.75	2295347.00	5886414.00	1607622209.00	15.824	307.617
Share capital	6622	3400133.46	7710776.71	157600.00	875460.00	1490550.00	3005579.00	105596201.00	7.849	78.689
Total capital	6622	1191068.29	5305519.35	-207088.00	59116.25	305537.50	879875.25	105919701.00	14.138	236.684
Finance costs	6622	109532.60	444821.87	0.00	4889.25	18172.50	57624.00	9556030.00	10.997	157.611
Current ratio	6622	359.3196	1518.8762	5.88	135.6825	183.7050	285.4175	71149.48	27.090	973.155
Quick ratio	6622	238.5410	789.1300	0.43	68.2750	114.3650	197.5925	18744.78	14.626	262.141
Interest expense ratio (B)	6622	3.6716	77.2482	0.00	0.16	0.56	1.36	5851.14	67.100	4980.851
Total liabilities / total net worth	6622	120.8301	543.0664	0.48	42.7750	80.3150	134.2325	41648.92	68.020	5168.925
Operating profit / paid-in capital ratio	6622	17.1705	40.4674	-777.80	-1.21	9.09	28.0125	631.99	2.509	58.180
Retention ratio	6622	15.4171	776.7598	-59557.14	21.8550	37.2850	54.1875	100.00	-69.230	5240.150

Missing Values

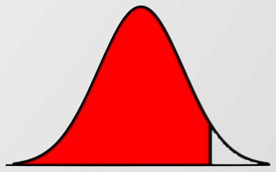
Imputed

Outlier Detection

|z| > 3

Descriptive Statistics

High skewness and kurtosis



TCRI Distribution

TCRI Rating	Count	Percentage (%)
5	2054	31.01
6	2161	32.62
7	1389	20.97
8	667	10.07
9	292	4.41
D	59	0.89
C	2	0.03

High frequency

5 and 6

Moderate to high risk

5-9, C, D

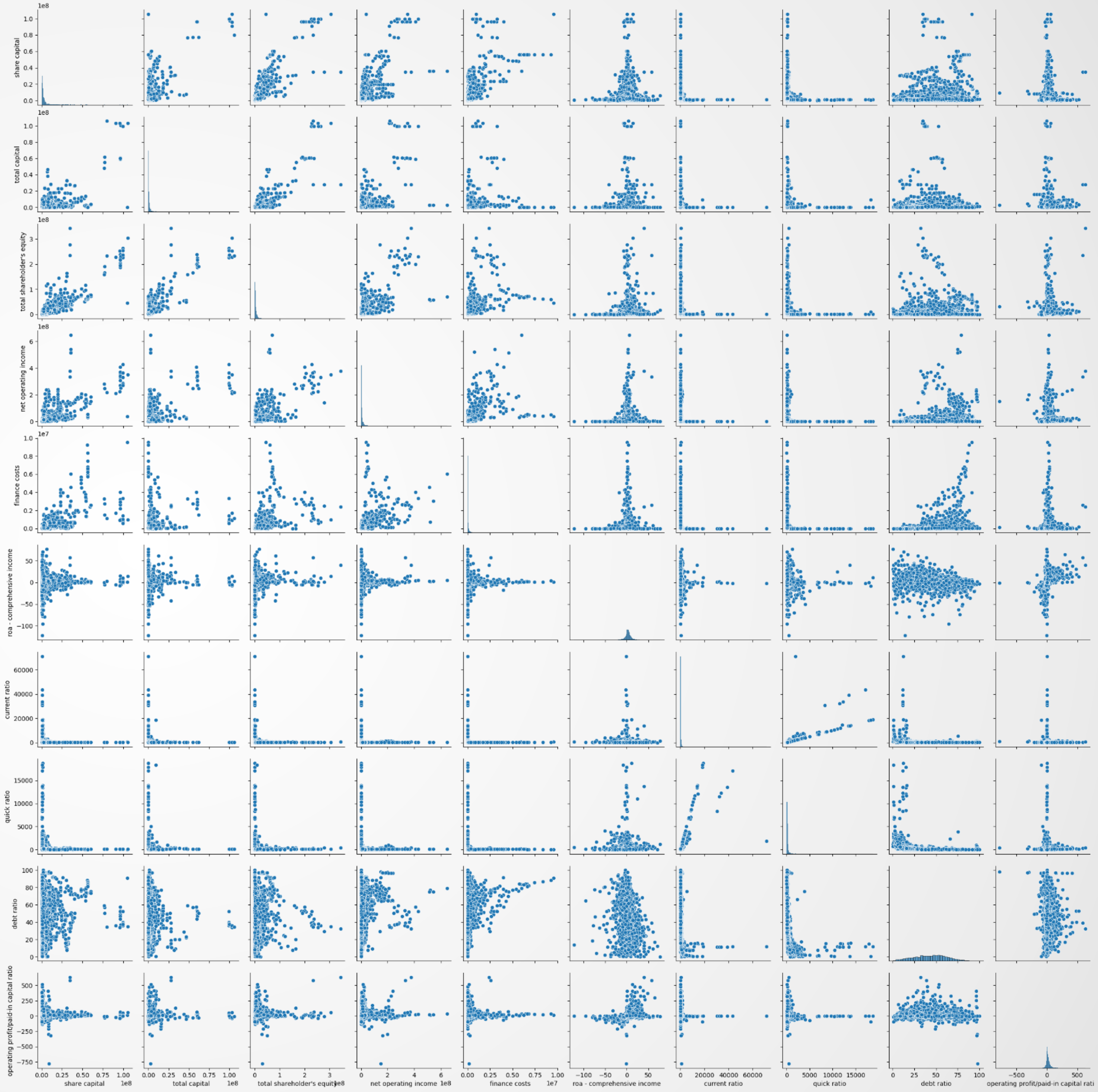
Data Imbalance

9, C, D

TCRI Visualization: Pairwise Scatter Plot

Untransformed form

10 predictors



TCRI Visualization: Heatmap

All ratios

Low to no Correlation

Size and cost

med to high correlation

Feature Selection Results

Model	Accuracy	Weighted Precision	Weighted Recall	Weighted F1
MLR: All predictors	0.55	0.54	0.55	0.54
MLR: Forward 10 preds	0.51	0.51	0.51	0.50
MLR: Forward 20 preds	0.52	0.51	0.52	0.51
MLR: Forward auto	0.52	0.51	0.52	0.51
MLR: Backward 10 preds	0.55	0.54	0.55	0.54
MLR: Backward 20 preds	0.54	0.53	0.54	0.53
MLR: Backward auto	0.54	0.54	0.54	0.53

Superior Method

Backward selection removes redundancy better

Accuracy

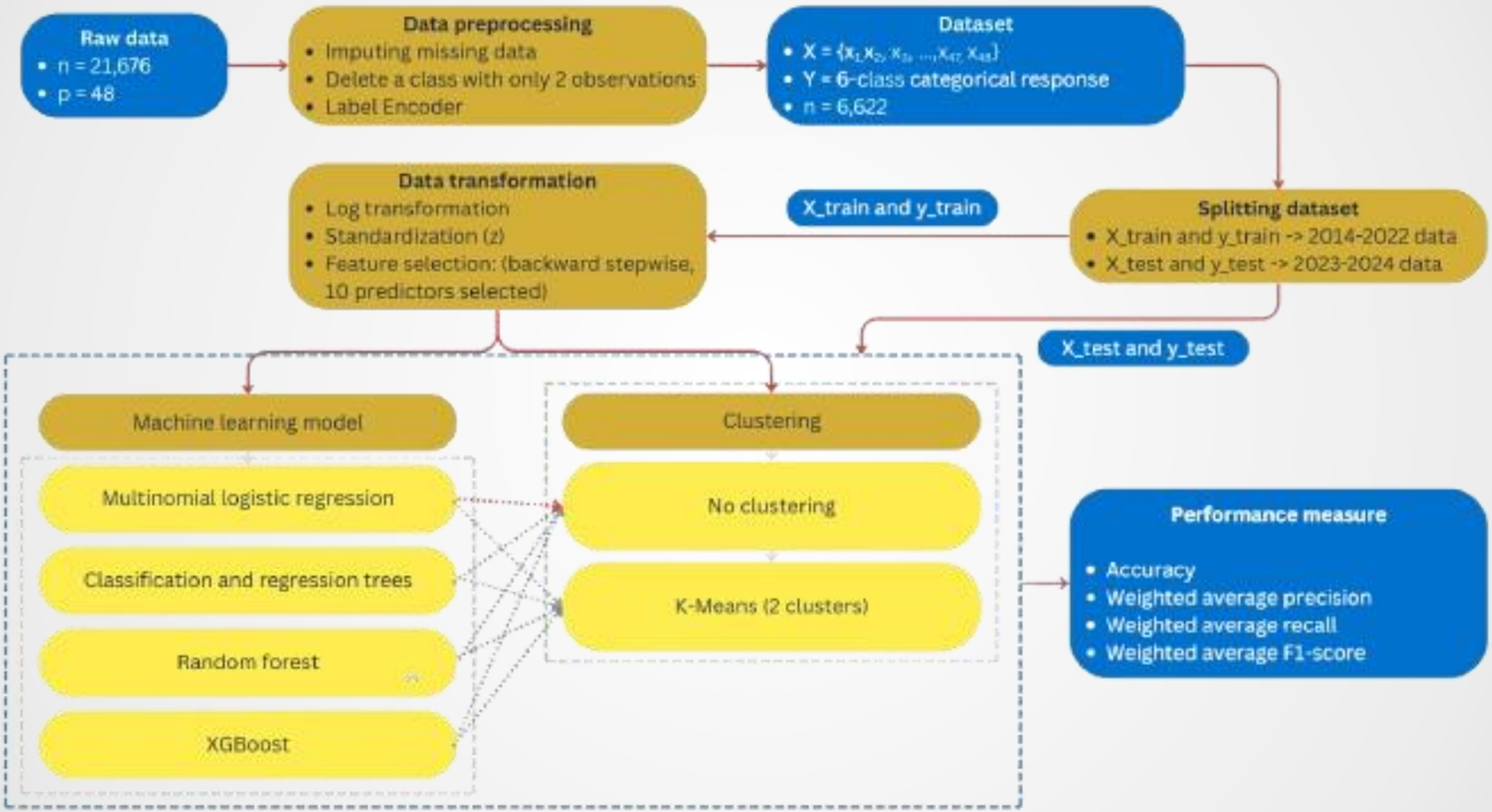
The model achieved **0.55** accuracy using only 10 selected predictors

Key Indicators

- Total liabilities
- Share capital
- Total capital
- Finance costs
- Current ratio
- Quick ratio
- Interest expense ratio (B)
- Total liabilities/total net worth
- Operating profit/paid-in capital ratio
- Retention ratio

Study Plan and Results

Study plan and flow chart



Model Performance

Ensemble Superiority
XGB and RF
Outperforming and capturing latent structure effectively

Clustering Impact
Negligible
In predictive accuracy for this specific TCRI dataset

Best performing model
XGB
0.63 prediction accuracy

Group	Model	Accuracy	Weighted Precision	Weighted Recall	Weighted F1
No cluster	Multinomial Logistic Regression	0.54	0.54	0.54	0.53
No cluster	CART	0.48	0.48	0.48	0.48
No cluster	RandomForest	0.62	0.62	0.62	0.62
No cluster	XGBoost	0.63	0.63	0.63	0.63
Cluster 1	Multinomial Logistic Regression	0.58	0.57	0.58	0.57
Cluster 1	CART	0.49	0.50	0.49	0.49
Cluster 1	RandomForest	0.60	0.60	0.60	0.60
Cluster 1	XGBoost	0.63	0.63	0.63	0.63
Cluster 2	Multinomial Logistic Regression	0.53	0.52	0.53	0.51
Cluster 2	CART	0.49	0.50	0.49	0.49
Cluster 2	RandomForest	0.61	0.62	0.61	0.61
Cluster 2	XGBoost	0.62	0.62	0.62	0.62

Conclusions

Findings

- XGBoost achieved the highest accuracy (~0.63).
- K-means pre-clustering added no benefit.
- Backward feature selection distilled predictors to 10 meaningful variables.

Implications

- Streamlined pipeline: ensemble methods + efficient feature engineering.
- Reject pre-clustering to save computation.
- Balance accuracy, interpretability, and resource efficiency.

Future Research

- Test alternative segmentation (Hierarchical Clustering, DBSCAN).
- Explore interpretable models (GAMs, MLPs).
- Advance to LightGBM/CatBoost ensembles.
- Integrate Explainable AI (SHAP) for regulatory transparency.

Thank you